Anomalous Precession of the Perihelion of Mercury

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Context

The conceptual structure of this derivation follows material introduced in advanced coursework at the Universitat de Barcelona. All detailed calculations, formal development, and typesetting are the independent work of the author.

Introduction

We employ the first integrals of motion associated with the Killing vectors of the Schwarzschild metric to study the anomalous precession of Mercury's perihelion—a foundational test of General Relativity that precisely accounts for the $43''$ (arcseconds) per century shift unexplained by Newtonian mechanics.

1. The Effective Radial Potential

Starting from the effective radial potential for a massive particle, which incorporates the standard Newtonian terms along with the General Relativistic correction ($\sim 1/r^3$), we evaluate the critical points by setting the radial derivative to zero:

$$V_{\text{eff}}(r) = -\frac{GM}{r} + \frac{l^2}{2r^2} - \frac{GMl^2}{c^2r^3}$$

$$\implies \frac{dV_{\text{eff}}(r)}{dr} = \frac{GM}{r^2} - \frac{l^2}{r^3} + \frac{3GMl^2}{c^2r^4}$$

Equating the derivative to zero to find the extrema:

$$GMc^2r^2 - l^2c^2r + 3GMl^2 = 0 \implies r^2 - \frac{l^2}{GM}r - 3\frac{l^2}{c^2} = 0$$

$$\implies r = \frac{1}{2}\frac{l^2}{GM} \pm \frac{1}{2}\sqrt{\frac{l^4}{G^2M^2} + 12l^2} = \frac{l^2}{2GM}\left(1 \pm \sqrt{1 + 12\frac{G^2M^2}{l^2c^2}}\right)$$

The solution with the negative sign is discarded as it yields an unphysical negative radius. Furthermore, the limit $r \to 0 \implies V_{\text{eff}} \to -\infty$, while $\lim_{r\to\infty} V_{\text{eff}} \to 0$. Thus, the physically acceptable positive root corresponds to a local maximum, differentiating it from the strictly repulsive Newtonian centrifugal barrier.

2. Perturbation Theory and the Orbital Equation

The energy equation of motion is given by:

$$E = \frac{1}{2}\dot{r}^2 + V_{\text{eff}}(r) = \frac{1}{2}\dot{r}^2 - \frac{GM}{r} + \frac{l^2}{2r^2} - \frac{GMl^2}{c^2r^3}$$

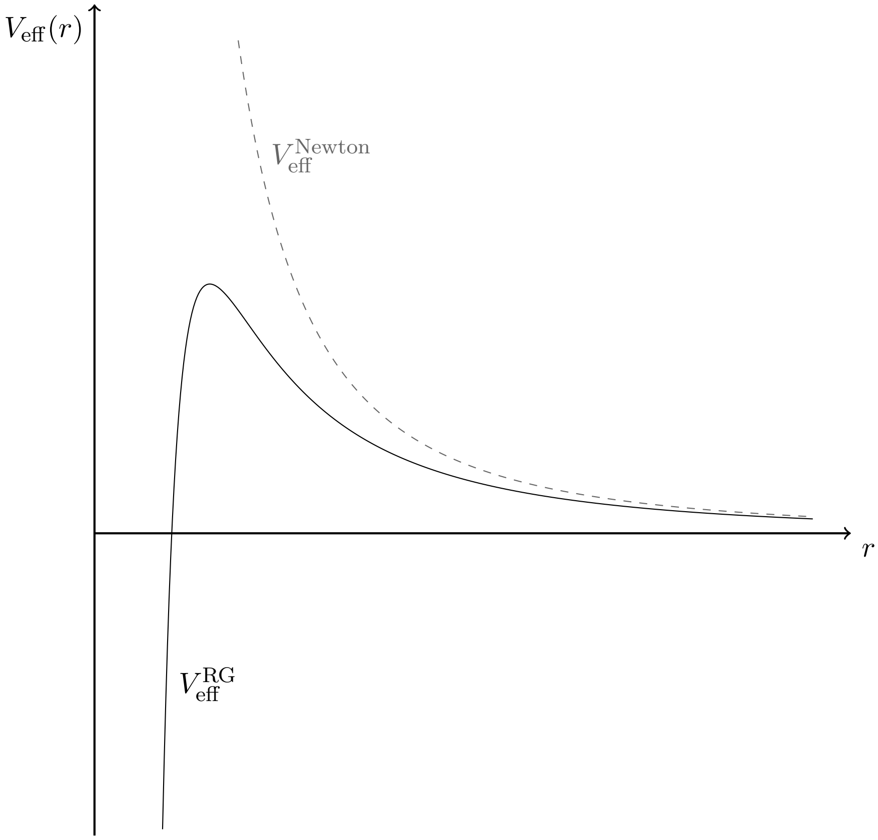

Figure 1: Comparison of the General Relativistic and Newtonian effective potentials.

We introduce the change of variable $u = \frac{1}{r}$. Denoting $u' \equiv \frac{du}{d\varphi}$ and utilizing the conserved angular momentum $r^2\dot{\varphi} = l$:

$$\dot{r} = \frac{dr}{du}\frac{du}{d\varphi}\frac{d\varphi}{dt} = -r^2 u'\dot{\varphi} = -lu'$$

$$\implies E = \frac{1}{2}u'^2 l^2 - GMu + \frac{l^2}{2}u^2 - \frac{GMl^2}{c^2}u^3$$

$$\implies \frac{2E}{l^2} = u'^2 - \frac{2GM}{l^2}u + u^2 - \frac{2GM}{c^2}u^3$$

We define the structural constants $C \equiv \frac{2E}{l^2}, p \equiv \frac{l^2}{GM}$, and the relativistic mass parameter $m \equiv \frac{GM}{c^2}$ to yield the non-linear differential equation:

$$u'^2 + u^2 - \frac{2}{p}u - 2mu^3 = C$$

As this equation lacks a closed-form analytic solution, we approximate the trajectory via perturbation theory. We decompose the solution as $u = u_0 + \epsilon u_1$, where $|u_1| \ll |u_0|$. Since m is small, we discard terms of order $\mathcal{O}(mu_1)$, and noting that u is extremely small for planetary orbits, we also discard $\mathcal{O}(u_0u_1)$ and $u_1^2 \to 0$. The expanded equation becomes:

$$C = (u_0' + u_1')^2 + (u_0 + u_1)^2 - \frac{2}{p}(u_0 + u_1) - 2m(u_0 + u_1)^3$$

$$= \left({u_0'}^2 + \cancel{{u_1'}^2} + 2u_0'u_1'\right) + \left(u_0^2 + \cancel{u_1^2} + \cancel{2u_0u_1}\right) - \frac{2}{p}\left(u_0 + u_1\right) - 2mu_0^3 - \cancel{\mathcal{O}(mu_1)}$$

$$= \left({u_0'}^2 + u_0^2 - \frac{2}{p}u_0 - C\right) + 2u_0'{u_1}' - \frac{2}{p}u_1 - 2mu_0^3 = 0$$

We isolate the unperturbed zeroth-order equation for u_0:

$$C = {u_0'}^2 + u_0^2 - \frac{2}{p}u_0$$

$$\implies 0 = \frac{1}{2{u_0}'}\frac{d}{d\varphi}\left({u_0'}^2 + u_0^2 - \frac{2}{p}u_0\right) = \frac{1}{2{u_0}'}\left(2u_0'u_0'' + 2u_0u_0' - \frac{2}{p}u_0'\right)$$

$$= {u_0}'' + u_0 - \frac{1}{p}$$

This reduces to the equation of a forced harmonic oscillator, whose general solution is:

$$u_0(\varphi) = D\cos(\varphi - \varphi_0) + \frac{1}{p}$$

where D and φ_0 are integration constants. Reorienting the axes such that $\varphi_0 = 0$, and defining $D = \frac{e}{p}$, we recover the classical Newtonian orbital geometry:

$$u_0 = \frac{1}{p}\left(1 + e\cos\varphi\right) \implies r = \frac{p}{1 + e\cos\varphi}$$

which defines an ellipse in polar coordinates, where e is the eccentricity and p is the semi-latus rectum.

3. Energy and Orbital Eccentricity

By substituting the unperturbed solution $u_0 = \frac{1}{p}(1+e\cos\varphi)$ (and its derivative $u_0' = -\frac{e}{p}\sin\varphi$) back into the non-linear constant equation, we determine the integration constant C:

$$C = \frac{e^2}{p^2}\sin^2\varphi + \frac{1}{p^2}\left(1 + e\cos\varphi\right)^2 - \frac{2}{p}\frac{1}{p}\left(1 + e\cos\varphi\right)^3$$

$$= \frac{1}{p^2}\left(e^2\sin^2\varphi + e^2\cos^2\varphi + 2e\cos\varphi + 1 - 2 - 2e\cos\varphi\right)$$

$$= \frac{1}{p^2}\left(e^2 - 1\right)$$

Recalling our initial definition $C \equiv \frac{2E}{l^2}$, the total specific energy of the particle is:

$$\frac{2E}{l^2} = \frac{1}{p^2}\left(e^2 - 1\right) \implies E = \frac{l^2}{2p^2}\left(e^2 - 1\right)$$

This rigorously confirms the classical mechanical bound state condition:

$$E < 0 \iff e^2 - 1 < 0 \iff e < 1$$

proving that for negative total energy, the closed orbit is strictly elliptical.

4. The Anomalous Precession Shift

Integrating the relativistic correction $\mathcal{O}(mu_0^3)$ leads to an approximated orbital equation defined by:

$$r(\varphi) = \frac{p}{1 + e\cos\varphi + \varepsilon e\varphi \sin\varphi}$$

Utilizing the trigonometric identity for the cosine of a sum, $\cos(A-B) = \cos A \cos B + \sin A \sin B$, and taking the first-order Taylor expansion for the small angle $\varepsilon\varphi \ll 1$:

$$e\cos([1-\varepsilon]\varphi) = e\cos\varphi\cos(\varepsilon\varphi) + e\sin\varphi\sin(\varepsilon\varphi) \simeq e\cos\varphi + e\sin\varphi(\varphi\varepsilon)$$

$$\implies r(\varphi) \simeq \frac{p}{1 + e\cos([1-\varepsilon]\varphi)}$$

Here, the anomalous term dictating the orbital precession is $\varepsilon = \frac{3m}{p}$. Expanding the semi-latus rectum as $p = a(1-e^2)$, where a is the semi-major axis, we evaluate ε using Mercury's astronomical parameters:

$$\varepsilon = \frac{3GM}{ac^2(1-e^2)} \approx 8 \times 10^{-8} \sim 10^{-7}$$

Evaluating the precession radially per revolution yields $\delta\varphi = 0.103''$ (arcseconds).

Total Precession over One Century

Given that the orbital period of Mercury is 0.241 Earth years, the total accumulation of the perihelion shift per Earth century is:

$$\Delta\varphi = 100 \cdot \frac{1}{0.241} \cdot 0.103'' \approx 43'' \text{ per century}$$

This theoretical prediction provides an exact match to the observed perihelion precession unaccounted for by Newtonian perturbations from other planets.